

JavaScript Math Reference

[Back to JS page](#)

Table of Contents

- [JavaScript Math Reference](#)
 - [Math Properties](#)
 - [Example 1](#)
 - [Math Rounding Methods](#)
 - [Example 2](#)
 - [Math Power and Root Methods](#)
 - [Example 3](#)
 - [Math Trigonometry Methods](#)
 - [Example 4](#)
 - [Math Utility Methods](#)
 - [Example 5](#)
 - [Document](#)
 - [Reference](#)

Math Properties

The complete `Math` object reference - properties (constants):

Property	Description
<code>Math.E</code>	Euler's number (~2.718)
<code>Math.PI</code>	Pi (~3.14159)
<code>Math.SQRT2</code>	Square root of 2 (~1.414)
<code>Math.SQRT1_2</code>	Square root of 1/2 (~0.707)
<code>Math.LN2</code>	Natural log of 2 (~0.693)
<code>Math.LN10</code>	Natural log of 10 (~2.303)
<code>Math.LOG2E</code>	Log base 2 of E (~1.443)
<code>Math.LOG10E</code>	Log base 10 of E (~0.434)

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Reference</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math Properties</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "Math.E: " + Math.E + "<br>" +
  "Math.PI: " + Math.PI + "<br>" +
  "Math.SQRT2: " + Math.SQRT2 + "<br>" +
  "Math.SQRT1_2: " + Math.SQRT1_2 + "<br>" +
  "Math.LN2: " + Math.LN2 + "<br>" +
  "Math.LN10: " + Math.LN10 + "<br>" +
  "Math.LOG2E: " + Math.LOG2E + "<br>" +
  "Math.LOG10E: " + Math.LOG10E;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

Math Rounding Methods

Reference for rounding methods:

Method	Description
<code>Math.round(x)</code>	Rounds to nearest integer
<code>Math.ceil(x)</code>	Rounds up to nearest integer
<code>Math.floor(x)</code>	Rounds down to nearest integer
<code>Math.trunc(x)</code>	Removes decimal part (no rounding)
<code>Math.sign(x)</code>	Returns -1, 0, or 1

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Reference</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math Rounding Methods</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "Math.round(4.6): " + Math.round(4.6) + "<br>" +
  "Math.ceil(4.2): " + Math.ceil(4.2) + "<br>" +
  "Math.floor(4.9): " + Math.floor(4.9) + "<br>" +
  "Math.trunc(4.7): " + Math.trunc(4.7) + "<br>" +
  "Math.sign(-5): " + Math.sign(-5) + "<br>" +
  "Math.sign(0): " + Math.sign(0) + "<br>" +
  "Math.sign(8): " + Math.sign(8);
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Math Power and Root Methods

Reference for power and root methods:

Method	Description
<code>Math.pow(x, y)</code>	x raised to the power y
<code>Math.sqrt(x)</code>	Square root of x
<code>Math.cbrt(x)</code>	Cube root of x
<code>Math.exp(x)</code>	e raised to the power x
<code>Math.log(x)</code>	Natural log of x
<code>Math.log2(x)</code>	Log base 2 of x
<code>Math.log10(x)</code>	Log base 10 of x

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Reference</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math Power and Root Methods</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "Math.pow(2, 3): " + Math.pow(2, 3) + "<br>" +
  "Math.sqrt(16): " + Math.sqrt(16) + "<br>" +
  "Math.cbrt(8): " + Math.cbrt(8) + "<br>" +
  "Math.exp(1): " + Math.exp(1).toFixed(4) + "<br>" +
  "Math.log(10): " + Math.log(10).toFixed(4) + "<br>" +
  "Math.log2(8): " + Math.log2(8) + "<br>" +
  "Math.log10(1000): " + Math.log10(1000);
</script>

</body>
</html>

```

Example 3

Result [View Example](#)

Math Trigonometry Methods

Reference for trigonometry methods:

Method	Description
<code>Math.sin(x)</code>	Sine of x (radians)
<code>Math.cos(x)</code>	Cosine of x (radians)
<code>Math.tan(x)</code>	Tangent of x (radians)
<code>Math.asin(x)</code>	Arcsine of x
<code>Math.acos(x)</code>	Arccosine of x
<code>Math.atan(x)</code>	Arctangent of x

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Reference</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math Trigonometry Methods</h4>
<p id="demo"></p>

<script>
// Angle in radians: 90 degrees = PI/2
const angle = Math.PI / 2;

document.getElementById("demo").innerHTML =
  "sin(PI/2): " + Math.sin(angle).toFixed(2) + "<br>" +
  "cos(0): " + Math.cos(0).toFixed(2) + "<br>" +
  "tan(PI/4): " + Math.tan(Math.PI / 4).toFixed(2) + "<br><br>" +
  "asin(1): " + Math.asin(1).toFixed(2) + " rad<br>" +
  "acos(1): " + Math.acos(1).toFixed(2) + " rad<br>" +
  "atan(1): " + Math.atan(1).toFixed(2) + " rad";
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

Math Utility Methods

Reference for utility methods:

Method	Description
<code>Math.abs(x)</code>	Absolute value of x
<code>Math.min(a, b, ...)</code>	Lowest value
<code>Math.max(a, b, ...)</code>	Highest value
<code>Math.random()</code>	Random number 0-1
<code>Math.hypot(a, b)</code>	Square root of ($a^2 + b^2$)
<code>Math.fround(x)</code>	Nearest 32-bit float

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Reference</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math Utility Methods</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "Math.abs(-7.5): " + Math.abs(-7.5) + "<br>" +
  "Math.min(3, 8, 1): " + Math.min(3, 8, 1) + "<br>" +
  "Math.max(3, 8, 1): " + Math.max(3, 8, 1) + "<br>" +
  "Math.random(): " + Math.random().toFixed(4) + "<br>" +
  "Math.hypot(3, 4): " + Math.hypot(3, 4) + " (3-4-5 triangle)<br>" +
  "Math.fround(1.5): " + Math.fround(1.5);
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Math Reference](#)